

EXPERIMENT 5

Student Name: Shagun

Branch: AIT-CSE

Semester: 5

Subject Name: Computer Network

UID: 24BDA70214

Section/Group: 24AIT_KRG-G2

Subject Code: 24CSH-337

EXPERIMENT 05

Title: Implement Dynamic Routing using RIP (Routing Information Protocol)

Aim

To create different network topologies using routers, switches, and PCs in Cisco Packet Tracer and configure dynamic routing using RIP so that devices from different networks can communicate successfully.

Objectives

- To understand the concept of routing.
- To understand the difference between static and dynamic routing.
- To understand the working of RIP.
- To assign IP addresses to network devices.
- To configure RIP on routers.
- To check routing tables.
- To test communication using Ping and Simple PDU.

Theory

Routing

Routing is the process of finding the best path for forwarding data packets from a source network to a destination network.

A router connects different networks and decides where the packet should be forwarded.

Example

PC1 ----- Router1 ----- Router2 ----- PC2

PC1 wants to send data to PC2. Since PC2 belongs to another network, the routers determine the path through which the packet should travel.

This process is called **Routing**.

EXPERIMENT 5

Why Do We Need Routing?

Routing is required for communication between different networks.

For example:

Network A : 192.168.1.0

Network B : 192.168.2.0

A router is required to forward packets between these networks.

Routing provides:

- **Inter-network Communication** – Connects different networks.
 - **Path Selection** – Finds a suitable path to the destination.
 - **Scalability** – Helps networks communicate as they grow.
 - **Internet Communication** – Allows communication between different networks.
-

Static Routing

Static routing is a method in which the network administrator manually adds routing information to the router.

The administrator specifies the destination network and the path through which packets should be forwarded.

Static routing is mainly suitable for small and stable networks.

Disadvantages of Static Routing

- **Manual Configuration:** Every route has to be configured manually.
 - **Not Scalable:** Managing many routes becomes difficult in large networks.
 - **Time Consuming:** More administrative work is required.
 - **No Automatic Updates:** Network changes require manual modification of routes.
 - **Configuration Errors:** Manual configuration can lead to incorrect routes.
-

Dynamic Routing

Dynamic routing is a method in which routers automatically learn and exchange routing information.

A routing protocol is used to exchange information between neighboring routers. The routers then automatically update their routing tables.

EXPERIMENT 5

Example

R1 ----- R2 ----- R3

R1 can learn about networks connected to R2 and R3 automatically through a dynamic routing protocol.

Therefore, the administrator does not need to manually configure every remote network.

Static Routing vs Dynamic Routing

Feature	Static Routing	Dynamic Routing
Configuration	Routes are manually configured	Routes are learned automatically
Route Updates	Must be updated manually	Updated automatically through routing protocols
Routing Protocol	Not required	Required
Administrative Work	More manual work	Less manual work
Scalability	Suitable for small networks	Suitable for networks with multiple routers
Network Changes	Administrator must modify routes	Routers can learn updated routes
Resource Usage	Uses less CPU and bandwidth	Uses CPU, memory and bandwidth for routing updates
Complexity	Simple for small networks	More complex but easier to manage as networks grow
Example	Static route	RIP, OSPF, EIGRP

Dynamic Routing Protocols

Some commonly used dynamic routing protocols are:

- **RIP** – Routing Information Protocol
- **OSPF** – Open Shortest Path First
- **EIGRP** – Enhanced Interior Gateway Routing Protocol
- **BGP** – Border Gateway Protocol

In this experiment, **RIP** is used for dynamic routing.

EXPERIMENT 5

RIP

RIP (Routing Information Protocol) is a **distance-vector routing protocol** that automatically exchanges routing information between routers.

RIP uses **Hop Count** as its routing metric to select a route.

How RIP Works

RIP routers exchange information about the networks they know with their neighboring routers.

For example:

R1 ----- R2 ----- R3

If R3 is connected to another network, R2 can learn about that network and share the information with R1. Thus, R1 can automatically learn about the remote network without manually adding a route.

Hop Count

A **hop** represents a router that a packet passes through while reaching the destination.

RIP uses the number of hops as its metric.

For example:

R1 ---- R2 ---- R3 ---- Network A

RIP uses the number of routers crossed to determine the route.

Important RIP Rules

- **1–15 hops:** Network is reachable.
- **16 hops:** Network is considered unreachable.
- RIP selects the route with the **lowest hop count**.
- Maximum usable hop count is **15**.

Example

Path 1: R1 → R2 → Network

Hop Count = 1

Path 2: R1 → R3 → R4 → Network

Hop Count = 2

RIP selects **Path 1** because it has a lower hop count.

EXPERIMENT 5

RIP Route Updates

RIP periodically exchanges routing information between neighboring routers.

The default RIP update interval is **30 seconds**.

The routers use these updates to learn new networks and update their routing tables.

RIP Convergence

Convergence is the process by which routers update their routing tables and reach a consistent view of the network after a change.

For example, if a link between two routers fails, the routers exchange updated routing information and modify their routing tables.

When all routers have updated information about the available paths, the network is said to be **converged**.

Advantages of RIP

- Easy to configure.
- Automatically learns routes.
- Requires less manual configuration.
- Suitable for small networks.
- Uses a simple routing metric.

Disadvantages of RIP

- Maximum usable hop count is 15.
 - Slow convergence compared with modern routing protocols.
 - Not suitable for large networks.
 - Uses only hop count and does not consider bandwidth or delay.
-

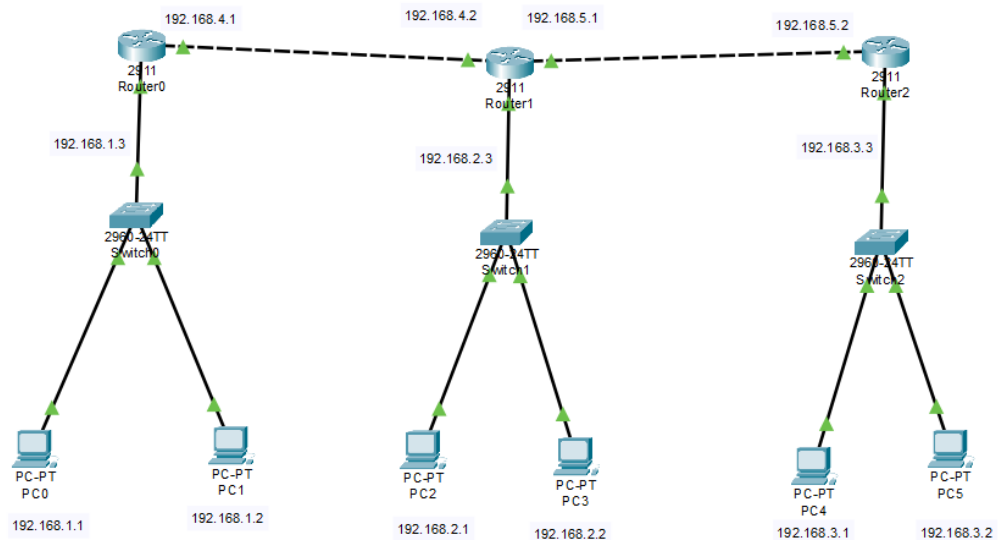
EXPERIMENT 5

Procedure

Part A – Network with 3 Routers, 3 Switches and 6 PCs

Steps

1. Open Cisco Packet Tracer.
2. Place 3 routers, 3 switches, and 6 PCs.
3. Connect two PCs to each switch.
4. Connect each switch to its corresponding router.
5. Connect Router0 to Router1 and Router1 to Router2.
6. Assign IP addresses to all PCs and router interfaces.
7. Configure RIP on all three routers.
8. Add the directly connected networks in the RIP configuration.
9. Save the configuration.
10. Check the routing table of the routers.
11. Verify communication using the **Ping** command.
12. Test the complete network using **Simple PDU**.



EXPERIMENT 5

Router0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Network

Network Address

192.168.1.0

192.168.4.0

Router1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

Network

Network Address

192.168.2.0

192.168.4.0

192.168.5.0

Router2

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

Network

Network Address

192.168.3.0

192.168.5.0

Router1

Gateway of last resort is not set

```

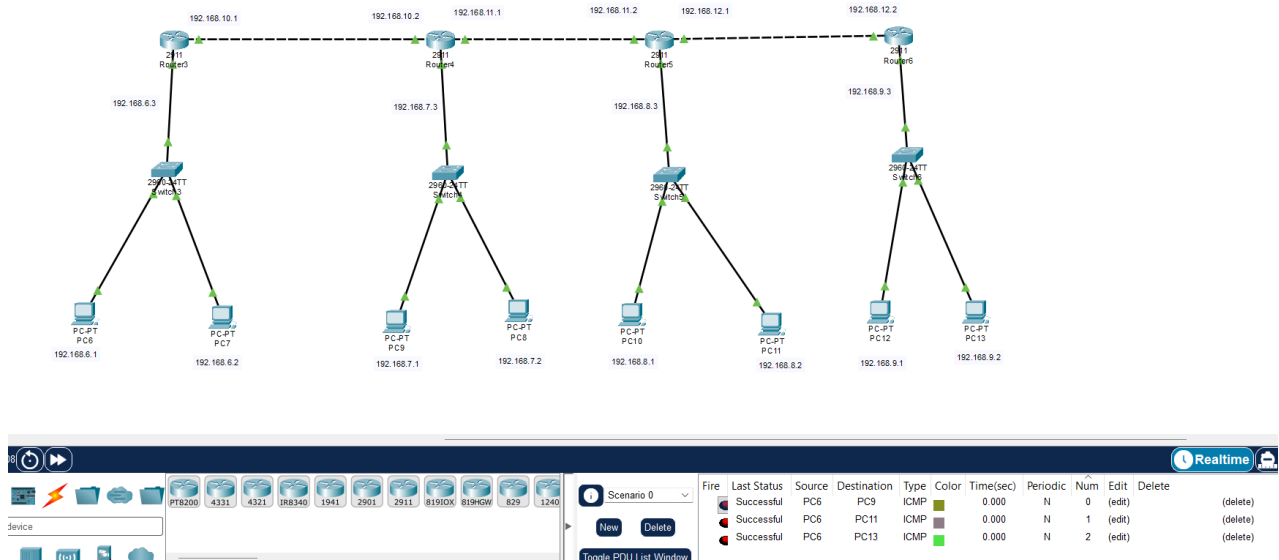
R    192.168.1.0/24 [120/1] via 192.168.4.1, 00:00:24, GigabitEthernet0/1
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, GigabitEthernet0/0
L    192.168.2.3/32 is directly connected, GigabitEthernet0/0
R    192.168.3.0/24 [120/1] via 192.168.5.2, 00:00:05, GigabitEthernet0/2
    192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.4.0/24 is directly connected, GigabitEthernet0/1
L    192.168.4.2/32 is directly connected, GigabitEthernet0/1
    192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.5.0/24 is directly connected, GigabitEthernet0/2
L    192.168.5.1/32 is directly connected, GigabitEthernet0/2
Router#

```

Part B – Network with 4 Routers, 4 Switches and 8 PCs

1. Open Cisco Packet Tracer.
2. Place 4 routers, 4 switches, and 8 PCs.
3. Connect two PCs to each switch.
4. Connect each switch to its corresponding router.
5. Connect the routers in series.
6. Assign IP addresses to all PCs and router interfaces.
7. Configure RIP on all four routers.
8. Add the directly connected networks in the RIP configuration.
9. Save the configuration.
10. Check the routing table of each router.
11. Verify connectivity using the **Ping** command.
12. Test communication between different networks using **Simple PDU**.

EXPERIMENT 5



Router3

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

Network

Network Address

192.168.6.0

192.168.10.0

Router4

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

Network

Network Address

192.168.7.0

192.168.10.0

192.168.11.0

Router5

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

Network

Network Address

192.168.8.0

192.168.11.0

192.168.12.0

Router6

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

Network

Network Address

192.168.9.0

192.168.12.0

Router5

Gateway of last resort is not set

```

R   192.168.6.0/24 [120/2] via 192.168.11.1, 00:00:21, GigabitEthernet0/1
R   192.168.7.0/24 [120/1] via 192.168.11.1, 00:00:21, GigabitEthernet0/1
R   192.168.8.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.8.0/24 is directly connected, GigabitEthernet0/0
L   192.168.8.3/32 is directly connected, GigabitEthernet0/0
R   192.168.9.0/24 [120/1] via 192.168.12.2, 00:00:12, GigabitEthernet0/2
R   192.168.10.0/24 [120/1] via 192.168.11.1, 00:00:21, GigabitEthernet0/1
R   192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.11.0/24 is directly connected, GigabitEthernet0/1
L   192.168.11.2/32 is directly connected, GigabitEthernet0/1
R   192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.12.0/24 is directly connected, GigabitEthernet0/2
--More--
  
```

☐ Top

EXPERIMENT 5

Result

Dynamic routing was successfully implemented in Cisco Packet Tracer using **RIP (Routing Information Protocol)** for different network topologies.

The routers automatically exchanged routing information and learned the remote networks. All devices were able to communicate successfully after configuring RIP. Connectivity was verified using the **Ping command and Simple PDU**.

Learning Outcomes

After completing this experiment, I was able to:

1. Understand the concept of routing.
2. Understand static and dynamic routing.
3. Understand the working of RIP.
4. Understand **hop count** as the RIP routing metric.
5. Configure IP addresses and router interfaces.
6. Configure RIP on multiple routers.
7. Check routing tables.
8. Understand RIP convergence.
9. Test communication using Ping and Simple PDU.